WORKED SOLUTION

STEP I 2004, Question 13

Probability and Statistics • Maximum of Uniforms

STEP I 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q13

The key idea

The maximum of several independent numbers is at most x exactly when *every* one of them is at most x. That turns the awkward "maximum" into a product of identical probabilities, giving the distribution function of X in one line. Everything else follows from it: differentiate for the density and the expectation, and set the distribution function equal to 0.05 to find the critical value of the test.

Solution

Part (i)

Let U_1, U_2, U_3 be the three numbers, independent and each uniform on $[0,1]$, so $\mathrm{P}(U_i \le x) = x$ for $0 \le x \le 1$. Since $X = \max(U_1, U_2, U_3)$,

$$X \le x \iff U_1 \le x,\ U_2 \le x \text{ and } U_3 \le x.$$

By independence,

$$F(x) = \mathrm{P}(X \le x) = x^3 \qquad (0 \le x \le 1).$$

In particular $\mathrm{P}(X \le 0.8) = 0.8^3 = 0.512$, as required.

Differentiating, X has density $f(x) = F'(x) = 3x^2$ on $[0,1]$, so

$$\mathrm{E}(X) = \int_0^1 x \cdot 3x^2 \,\mathrm{d}x = \left[\tfrac{3}{4}x^4\right]_0^1 = \frac{3}{4}.$$

This is sensible: the largest of three numbers should sit well above the mean $\frac{1}{2}$ of a single one.

Part (ii)

Now there are N independent numbers, each uniform on $[0, a]$, so $\mathrm{P}(U_i \le x) = x/a$ for $0 \le x \le a$, and the same argument gives

$$\mathrm{P}(X \le x) = \left(\frac{x}{a}\right)^N \qquad (0 \le x \le a),$$

with $\mathrm{P}(X \le x) = 1$ for $x \ge a$. Since X is continuous, $\mathrm{P}(X < x)$ is the same.

The critical value. The test rejects H_0 when $x < c$, and a 5% significance level means the probability of rejecting when H_0 is true ($a = 1$) is 0.05:

$$\mathrm{P}(X < c \mid a = 1) = c^N = 0.05, \qquad \text{so} \qquad c = 0.05^{1/N}.$$

Small values of x are evidence for $a < 1$, which is why the rejection region is $x < c$.

> **Ask yourself** How does "$x \le 0.8$ always leads to rejection" become an inequality in N?

Every $x \le 0.8$ is rejected exactly when every such x lies below c, that is, when $0.8 < c$. Since $t \mapsto t^N$ is increasing for $t > 0$, this is equivalent to

$$0.8^N < c^N = 0.05.$$

So we need the smallest integer N with $0.8^N < 0.05$. (Note that c grows towards 1 as N increases, so larger N makes the test more willing to reject.)

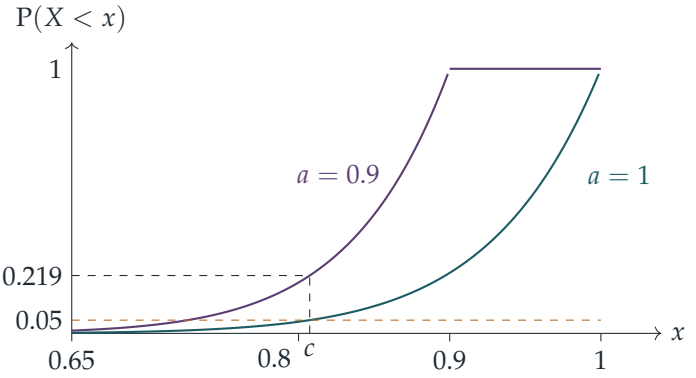
Using $2^{10} \approx 10^3$. Write $0.8 = 2^3/10$, so $0.8^N = 2^{3N}/10^N$.

- $N = 13$: $0.8^{13} = \dfrac{2^{39}}{10^{13}} = \dfrac{2^9 \cdot 2^{30}}{10^{13}} \approx \dfrac{512 \times 10^9}{10^{13}} = 0.0512 > 0.05.$
- $N = 14$: $0.8^{14} = \dfrac{2^{42}}{10^{14}} = \dfrac{2^2 \cdot 2^{40}}{10^{14}} \approx \dfrac{4 \times 10^{12}}{10^{14}} = 0.04 < 0.05.$

Since 0.8^N decreases as N increases, every $N \le 13$ also fails. So the smallest value is

$$N = 14.$$

(The approximation slightly underestimates powers of 2, since $2^{10} = 1024$; the true values $0.8^{13} \approx 0.0550$ and $0.8^{14} \approx 0.0440$ lead to the same conclusion.)



With $N = 14$, the critical value is $c = 0.05^{1/14} \approx 0.807$, just above 0.8.

If $a = 0.8$. All the numbers lie in $[0, 0.8]$, so certainly $X \leq 0.8 < c$. The null hypothesis is rejected with probability 1.

If $a = 0.9$. Now $c \approx 0.807 < 0.9$, so

$$\mathrm{P}(\text{reject}) = \mathrm{P}(X < c \mid a = 0.9) = \left(\frac{c}{0.9}\right)^{14} = \frac{c^{14}}{0.9^{14}} = \frac{0.05}{0.9^{14}}.$$

Here $0.9^2 = 0.81$, $0.9^4 = 0.6561$, $0.9^8 \approx 0.4305$, and so $0.9^{14} = 0.9^8 \cdot 0.9^4 \cdot 0.9^2 \approx 0.2288$. Therefore

$$\mathrm{P}(\text{reject} \mid a = 0.9) = \frac{0.05}{0.9^{14}} \approx 0.219.$$

So even with 14 observations, the test usually fails to detect that a is 0.9 rather than 1.

Checks

- A Monte Carlo simulation (2×10^5 trials) gave $\mathrm{P}(X \leq 0.8) \approx 0.512$ and $\mathrm{E}(X) \approx 0.750$ in part (i).
- In general, the maximum of n uniform numbers on $[0, 1]$ has mean $\frac{n}{n+1}$; for $n = 1$ this is $\frac{1}{2}$, as it should be, and for $n = 3$ it is $\frac{3}{4}$.
- With $N = 14$ and $c = 0.05^{1/14}$, simulation gave a rejection rate of about 0.0505 when $a = 1$ (the size of the test) and about 0.218 when $a = 0.9$, matching $0.05/0.9^{14} = 0.2186$.

Common slips

- Using $1 - 0.8^N$ in place of 0.8^N: the probability that the maximum is *small* is the product 0.8^N, not its complement.

- Taking the rejection probability at $a = 0.9$ to be $(0.8/0.9)^{14}$. The test rejects when $x < c$, and c is not 0.8.
- Stopping at $N = 13$ because 0.0512 "is about 0.05". We need strictly less than 0.05.

Beyond STEP

The maximum is an *order statistic*, and for the uniform distribution on $[0, a]$ it is a *sufficient statistic* for a: it contains all the information the sample holds about a. The rejection probability as a function of the true a is the *power function* of the test. Here it is $\min\{1,\ 0.05/a^N\}$ for $a \leq 1$, and comparing power functions is how statisticians choose between tests. Grimmett and Welsh, *Probability: An Introduction*, covers the distributions of maxima and minima; Blitzstein and Hwang, *Introduction to Probability*, treats order statistics in detail.

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